

Alvaro Javier Vargas Guerrero

PHD STUDENT · VUB

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Education

Vrije Universiteit Brussel

PHD ARTIFICIAL INTELLIGENCE

- Promotor: Prof. Dr. Guy Nagels
- Promotor: Prof. Dr. Johan Loeckx

Brussels, Belgium

July 2025 - present

Vrije Universiteit Brussel

MASTER APPLIED SCIENCES AND ENGINEERING

- Promotor Master's Thesis: Prof. Dr. Ann Nowé
- Promotor Master's Thesis: Prof. Dr. Guy Nagels
- Advisor Master's Thesis: Dr. Stijn Denissen

Brussels, Belgium

September 2022 - July 2024

Professional Experience

2024-2025 **IT Consultant**, Addestino

2022-2023 **Software Developer**, Vrije Universiteit Brussel

Publications

PUBLISHED

Vargas Guerrero, A. J., Wang, X., & Doan, Q. M., & Nagels, G. *Beyond Weights and Gradients: A Taxonomy of Federated Learning Messages.*

Wang, X., Nguyen, T. L., Chén, O. Y., Vargas Guerrero, A. J., Doan, Q. M., Hu, T., Denissen, S., Van Schependom, J., & Nagels, G. *Federated Adversarial Network (FAN): A Generative Proxy-Based, Defense-Enhanced, and Non-IID-Robust Federated Learning Method.*

IN REVIEW

Vargas Guerrero, A. J., Dillen, A., Loeckx, J., & Nagels, G. *A Systematic Review of Federated Structure Learning: Definitions and Methods.*

Teaching Experience

2026 **Statistical Foundations of Machine Learning**, Teaching Assistant

2026 **Arts in de Maatschappij**, Teaching Assistant

2026 **Current Trends in Artificial Intelligence**, Teaching Assistant

Mentoring

2026 **Kamila Maria Obregon Ortega**, Internship on Federated Causal Discovery

2026 **Yaun Francis**, Internship on Federated Causal Discovery

2027 **Mohammad Amman Ul Haq Gadda**, Master Thesis on Medical Federated Causality

2027 **Houda Yousfi**, Master Thesis on Federated Causal Discovery

2027 **Srivas Hari**, Master Thesis on Federated Causal Discovery